Predicting Interstitial ELF and Mapping Void Connectivity for Electride Screening

A graph neural network predicts the electron localization function (ELF) at interstitial voids; an exact geometric network labels how those voids connect

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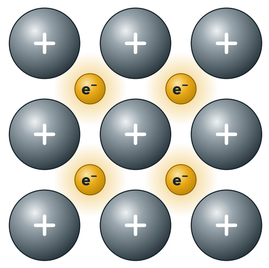
GitHub repository

1 BACKGROUND

WHAT IS AN ELECTRIDE?

Electrons that sit between the atoms

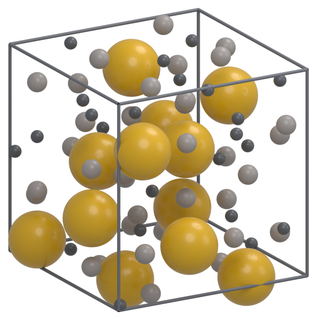
In an electride, some valence electrons are not bound to any atom. They localize in the empty space (voids) of the crystal and act as the anions.



Schematic: cations (+) with electrons (e−) held in the voids between them.

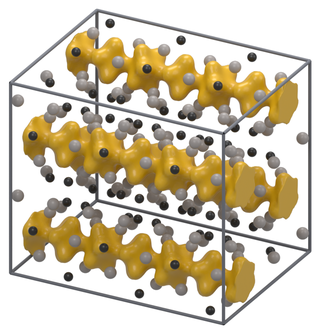
Those voids can be isolated or connected, which sets the dimensionality

0D isolated cages



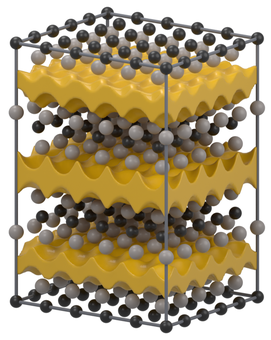
$Ca_6Al_7O_{16}$ (mayenite type)

1D channels



Y_5Si_3

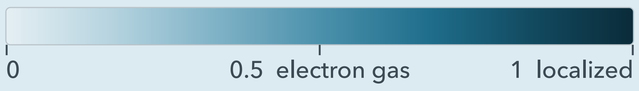
2D sheets



Ca_2N

How electrides are identified: ELF

The electron localization function (ELF) runs from 0 to 1 and measures how localized electrons are compared with a uniform electron gas (0.5).



An ELF maximum inside a void, away from any nucleus, signals interstitial electrons.

The bottleneck: ELF needs a DFT calculation for every candidate crystal.

2 GOAL

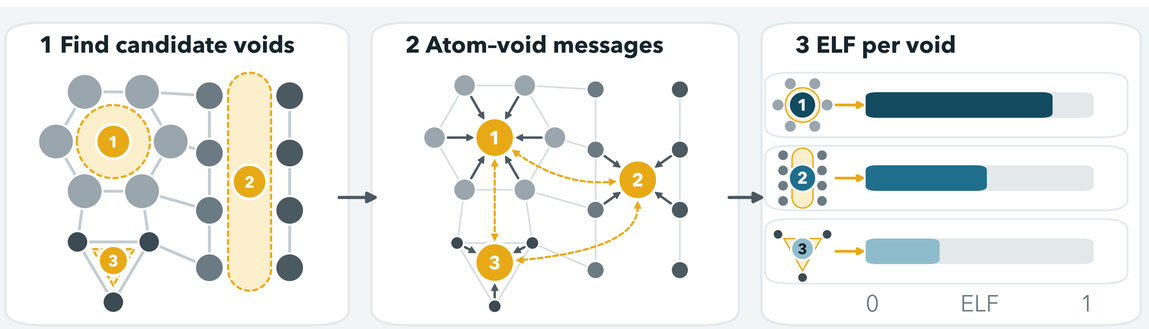
Screen crystals for electrides **from structure alone**: predict ELF at each void without DFT, and map how the voids connect. We built two void-centered tools and tested each one separately.

3 TOOL 1 · PREDICT ELF AT EACH VOID

GRAPH NEURAL NETWORK

Can a graph network predict the ELF at each candidate void from the surrounding atoms?

Atoms and candidate voids are both nodes of one periodic graph, joined by distance edges. Message passing gathers each void's atomic neighborhood and returns one ELF value at the void center.



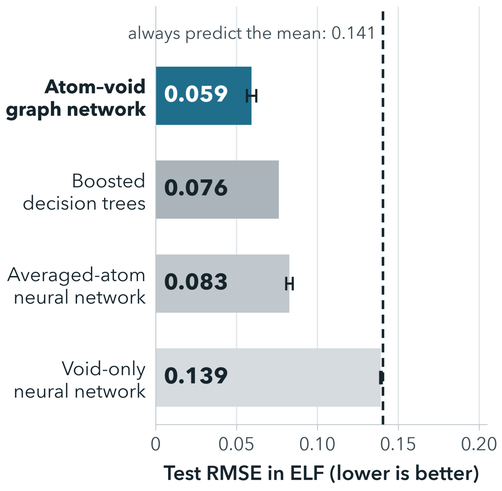
0.059 test RMSE (± 0.003) on the 0–1 ELF scale

0.82 test R^2

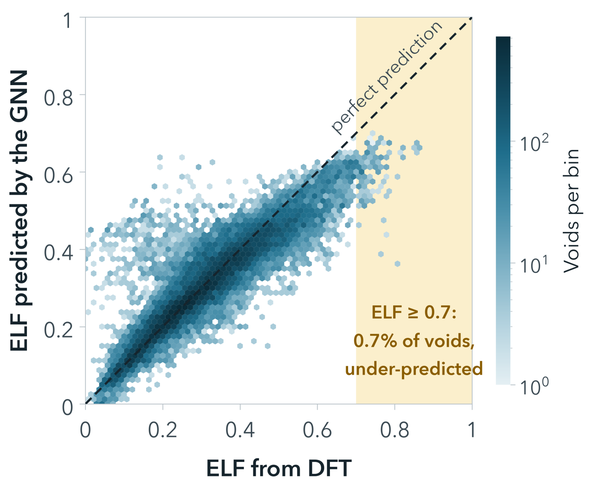
88,680 test voids in 5,000 crystals

9 structure families held out for the test

Graph network vs. simpler models



Predicted vs. DFT ELF on the test set



The atoms carry the signal. A void's own geometry does no better than predicting the mean (0.139 vs 0.141); adding its atomic neighborhood cuts the error by more than half.

Data: ELFNet [2], 50,000 DFT structures (metal frameworks derived from compressed hydrides), split by structure family.

4 TOOL 2 · MAP HOW THE VOIDS CONNECT

EXACT GEOMETRY

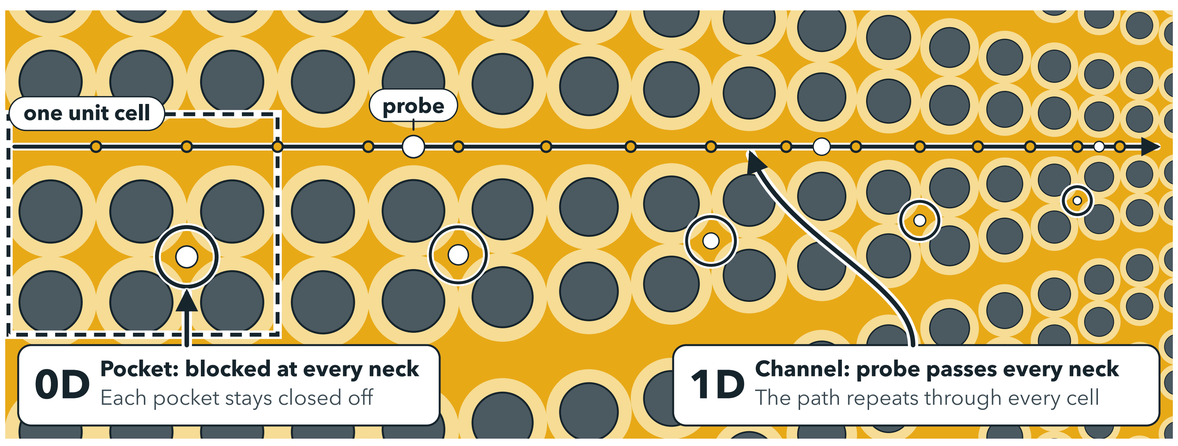
Can geometry alone recover the dimensionality reported for electrides?

Each atom blocks a sphere (ionic radius + probe radius). An exact power-diagram network of the open space that remains counts the directions in which it repeats through the crystal.

Probe must fit to pass through a neck

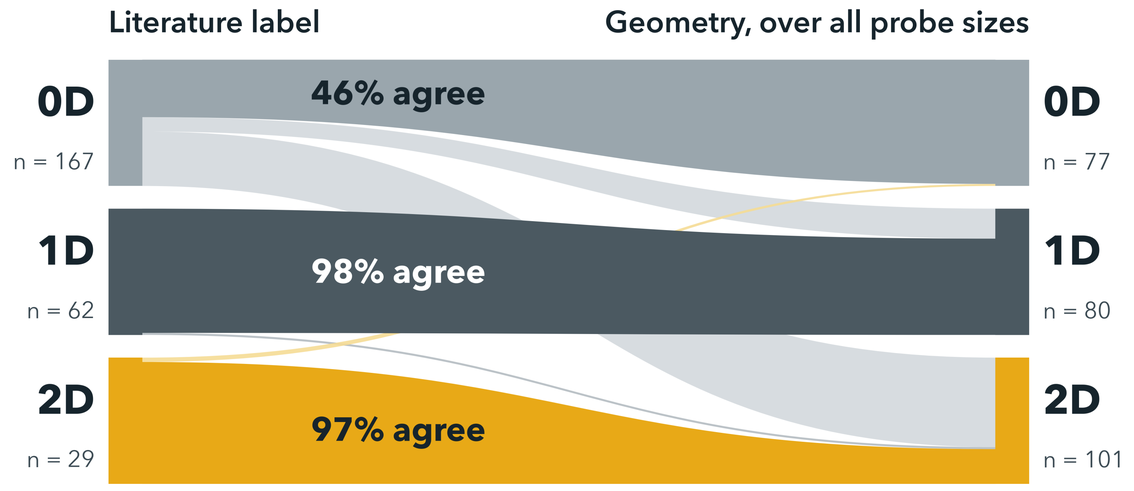
Atom + probe radius blocked: probe center cannot enter

Open space where the probe can sit



Probe-size sweep

EXPLORATORY



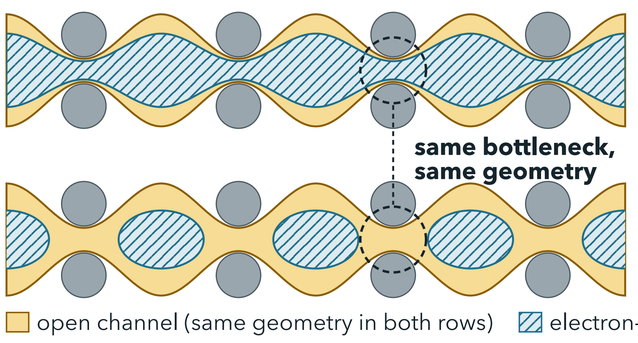
In 91 of 167 electrides labeled 0D, the open space connects at some probe size. The space forms a path, but the electrons may still sit in separate pockets.

5 PROPOSED · COMBINE THE TWO TOOLS

FUTURE WORK, NOT YET TESTED

Open space is not the same as electrons

The same open channel can hold connected electrons (1D) or separate pockets (0D); only the ELF at its bottleneck tells them apart.



Connected electron-rich region
High ELF at the bottleneck
Electrons connect: 1D

Separate electron-rich pockets
Low ELF at the bottleneck
Electrons stay apart: 0D

TOOL 2 · GEOMETRY
Find the channel and its narrowest bottleneck

↓

TOOL 1 · GNN
Predict the ELF at that bottleneck

↓

COMBINED
Structural vs. electronic dimensionality

6 TAKEAWAYS

- **Void ELF is predictable from nearby atoms:** RMSE 0.059, R^2 0.82 on held-out structure families.
- **Geometry finds the 1D and 2D electrides:** the sweep agrees with the literature label for 98% of 1D and 97% of 2D.
- **Open space alone cannot call 0D:** 91 of 167 electrides labeled 0D still have connected open space at some probe size.

NEXT STEPS

- Run DFT on known electrides and test the GNN on them.
- Predict ELF at bottlenecks, not only at void centers.
- Combine both tools into one crystal-level electride screen.

References [1] Becke & Edgecombe, J. Chem. Phys. 92, 5397 (1990), ELF. [2] Ellis, Ellis & Miao, J. Mater. Inf. 6, 45 (2026), doi:10.20517/jmi.2026.44, ELFNet. [3] Registry labels: Magnetic electrides, J. Am. Chem. Soc. (2023), doi:10.1021/jacs.3c00284; Zhu, Frolov & Choudhary, Matter 1, 1293 (2019); Burton et al., Chem. Mater. 30, 7521 (2018). [4] Methods: Shannon, Acta Cryst. A32, 751 (1976), ionic radii; Willems et al., Micropor. Mesopor. Mater. 149, 134 (2012), Zeo++; Gilmer et al., ICML (2017), message passing. [5] Related work: Xiao, Bristowe & Mostofi, Theory and Discovery of Electrides, arXiv:2605.11724 (2026); Qin, Lv & Zhong, EAC-Net: Predicting Real-Space Charge Density via Equivariant Atomic Contributions, arXiv:2508.04052; Ryan, Lengyel & Shatruk, Crystal Structure Prediction via Deep Learning, J. Am. Chem. Soc. 140, 10158 (2018); Wang et al., Machine Learning-Accelerated Discovery of A_2BC_2 Ternary Electrides with Diverse Anionic Electron Densities, J. Am. Chem. Soc. 145, 26412 (2023). **Acknowledgements** [advisor, funding]; Purdue RCAC (Negishi).